ISML – Hosting & AI Infrastructure Cost Analysis

Infrastructure Requirements & Estimated Monthly Expenses

1. Infrastructure Cost Comparison

The following table presents the estimated monthly infrastructure expenses for ISML at different student capacities. Calculations account for core application compute, resilient multi-tenant database clusters, global content delivery, low-latency WebRTC live broadcast, and managed intelligent conversational speech agents.

Infrastructure	500 Students	5,000 Students	5,00,000 Students
Frontend & Cloudflare CDN	₹1,660	₹12,500	₹3,75,000
Backend API Hosting	₹4,150	₹29,000	₹11,60,000
Supabase Database + Redis	₹3,750	₹25,000	₹10,40,000
Cloudflare R2 Storage	₹1,250	₹6,600	₹2,90,000
LiveKit – Live Classes	₹5,800	₹50,000	₹31,50,000
Cloudflare Security	₹1,660	₹16,600	₹2,50,000
AI Agents – Resources & Exams	₹2,500	₹20,800	₹15,00,000
AI Speaking Agent – Estimated Budget	₹62,500	₹6,25,000	₹5,15,00,000
DevOps & Monitoring	₹2,080	₹10,000	₹3,30,000
Estimated Infrastructure Subtotal	**₹85,350**	**₹7,95,500**	**₹5,95,95,000**
Gemini API Usage	**Usage-based**	**Usage-based**	**Usage-based**
Final Monthly Total	**Subtotal + Gemini API**	**Subtotal + Gemini API**	**Subtotal + Gemini API**

The infrastructure subtotal is calculated from the supplied estimates. Gemini API charges must be measured separately unless they are already included in the AI budget estimates. These figures are planning estimates, not confirmed vendor quotations.

Per-Student Monthly Cost Breakdown (in INR ₹)

The following table breaks down the exact infrastructure and AI expenses calculated on a per-student monthly basis across all three student capacities. This clearly demonstrates institutional economies of scale as learner density expands.

Infrastructure Component	Cost / Student (500 Scale)	Cost / Student (5,000 Scale)	Cost / Student (5,00,000 Scale)
Frontend & Cloudflare CDN	₹3.32	₹2.50	₹0.75
Backend API Hosting	₹8.30	₹5.80	₹2.32
Supabase Database + Redis	₹7.50	₹5.00	₹2.08
Cloudflare R2 Storage	₹2.50	₹1.32	₹0.58
LiveKit – Live Classes	₹11.60	₹10.00	₹6.30
Cloudflare Security	₹3.32	₹3.32	₹0.50
AI Agents – Resources & Exams	₹5.00	₹4.16	₹3.00
AI Speaking Agent – Estimated Budget	₹125.00	₹125.00	₹103.00
DevOps & Monitoring	₹4.16	₹2.00	₹0.66
Total Cost Per Student / Month	**₹170.70**	**₹159.10**	**₹119.19**

Per-student cost demonstrates significant cost reduction from ₹170.70 at pilot scale down to ₹119.19 at national scale, allowing ISML to achieve superior operating margins across institutions.

Gemini API – Token Pricing (in INR ₹)

ISML uses Gemini API to power AI-based learning features. Google charges according to the selected model and the amount of information processed and generated. All pricing below is converted and presented strictly in Indian Rupees (INR ₹ at ₹83.33/USD).

Example: Official Gemini 2.5 API standard pricing (in INR ₹)

Gemini Model	Input Cost per 1 Million Tokens	Output Cost per 1 Million Tokens
Gemini 2.5 Flash – Text	₹25.00	₹208.33
Gemini 2.5 Flash – Audio	₹83.33	Depends on output modality
Gemini 2.5 Flash-Lite – Text	₹8.33	₹33.33

Gemini 2.5 Flash-Lite – Audio	₹25.00	Depends on output modality

Audio input and text output are billed according to their respective token categories. Audio generation, if used, has separate pricing. Rates are presented in INR based on standard USD conversion; verify the selected model's current pricing before deployment. Official pricing reference: https://ai.google.dev/gemini-api/docs/pricing

How Student Usage Affects Gemini API Cost

The Speaking Agent processes student audio and generates individual feedback. More practice sessions and longer conversations generally result in higher API usage.

Student Usage Policy	Daily Practice Limit	Expected Cost Impact
Basic Access	2 drills per day	Lower API usage
Standard Access	3 drills per day	Moderate API usage
Extended Access	5 drills per day	Higher API usage
Unlimited Access	No daily limit	Potentially very high and unpredictable API usage

Important: These are proposed usage plans, not measured per-student costs. Actual charges depend on audio duration, input tokens, generated feedback length, model choice and retries.

2. Infrastructure – Purpose & Business Explanation

To provide full clarity for managerial and non-technical stakeholders, the following matrix explains the exact business purpose of each cloud technology component deployed within the ISML ecosystem.

Infrastructure Component	Why We Use It
Frontend – Next.js / CDN	Provides the ISML website and dashboards for students, teachers and management with faster page loading.
Backend – NestJS	Manages login, courses, batches, attendance, assessments and application operations.
Supabase PostgreSQL	Stores student profiles, college details, courses, academic records, attendance, payments and learning progress.
Redis Cache	Improves response speed for frequently accessed information and manages temporary data.
Cloudflare R2	Stores recorded classes, audio files, PDFs, presentations and learning

	materials.
LiveKit	Enables teachers and students to conduct live audio and video classes directly through ISML.
Cloudflare Security	Protects the platform against malicious traffic and common web attacks.
Gemini API	Provides the AI processing power for speaking evaluation, resource generation, examination evaluation and doubt solving. Charges depend on actual token and audio usage.
DevOps & Monitoring	Monitors platform performance, errors, availability and technical issues.

Overall: These services work together to manage ISML's academic operations, student information, live learning, recorded content and AI-powered learning services.